

Problem Set 5: Behavioral Economics — Deviations, Biases & Nudges

Microeconomic Theory for Experimentalists & Applied Microeconomists

Due: [date — before Lecture 1 of Module 6]

Ground rules: collaboration on ideas is encouraged; write-ups are individual. You may use AI tools for parts flagged with $\langle AI \rangle$ and must attach the transcript. Show all calculations.

Weights: Part A — 30 points. Part B — 45 points. Part C — 25 points.

Part A: Theory — Guided Calculations (30 points)

Question 1 (Present bias, sophistication, and the demand for commitment — 18 points). *An agent lives for three periods and must divide wealth $W = 12$ across consumption c_1, c_2, c_3 with utility $\ln c$ each period. She has β - δ preferences with $\beta = 0.7$ and $\delta = 1$: standing in any period, utility is*

$$\ln c_{\text{now}} + 0.7 \times (\text{sum of } \ln c \text{ in all remaining periods}).$$

- (a) **The exponential benchmark (3 pts).** *If $\beta = 1$, argue in two sentences (no calculus needed — think symmetry) that the agent consumes $(4, 4, 4)$.*
- (b) **The period-1 plan (4 pts).** *With $\beta = 0.7$, the period-1 self solves: maximize $\ln c_1 + 0.7 \ln c_2 + 0.7 \ln c_3$ subject to $c_1 + c_2 + c_3 = 12$. Show that the plan is $c_1 = 5$, $c_2 = c_3 = 3.5$. Hint: at the optimum, a dollar moved between any two periods must not raise utility, so $1/c_1 = 0.7/c_2 = 0.7/c_3$. Substitute into the budget.*
- (c) **The naif's period-2 betrayal (4 pts).** *Period 2 arrives with $12 - 5 = 7$ remaining. The period-2 self maximizes $\ln c_2 + 0.7 \ln c_3$ subject to $c_2 + c_3 = 7$. Show she chooses $c_2 \approx 4.12$, $c_3 \approx 2.88$ — not the $(3.5, 3.5)$ the period-1 self planned. State in one sentence what the naive agent got wrong.*
- (d) **Commitment demand as a diagnostic (5 pts).** *A bank offers a free account that lets the agent choose c_1 freely but locks the remainder into equal halves: $c_2 = c_3 = (12 - c_1)/2$.*
 - (i) *Compute the period-1 utility of the sophisticated agent with and without the account. You may take $c_1 = 5$ in both cases (with log utility, re-optimizing c_1 changes nothing); without the account she correctly foresees part (c). Show the account raises her utility.*
 - (ii) *Explain in two sentences why the naive agent's willingness to pay for this account is exactly zero.*
- (e) **Interpretation (2 pts).** *In one paragraph: what does this result let a field researcher measure by randomizing the price of a commitment account? Connect to the 28% take-up rate in Ashraf, Karlan & Yin (2006).*

Question 2 (Inequity aversion with numbers — 12 points). A responder in a \$10 ultimatum game has Fehr–Schmidt utility¹

$$U = x_{\text{self}} - \alpha \max(x_{\text{other}} - x_{\text{self}}, 0) - \gamma \max(x_{\text{self}} - x_{\text{other}}, 0).$$

- (a) (4 pts) Assume $\gamma < 1$ and consider offers $s \leq 5$. Show that the responder accepts an offer of s dollars if and only if $s \geq 10\alpha/(1 + 2\alpha)$. Compute the threshold for $\alpha = 0$, $\alpha = 0.5$, and $\alpha = 1.5$.
- (b) (4 pts) A population of responders has α uniformly distributed on $[0, 1.5]$. What fraction rejects an offer of \$2? Of \$3? Hint: invert the threshold formula to find the critical α for each offer.
- (c) (4 pts) In one paragraph: which experimental moments identify α and which identify γ ? (Think about which game puts the decision-maker behind, and which puts her ahead.) Why can't the ultimatum game alone pin down both?

Part B: Experimental Design (45 points)

Question 3 (Measuring the sophistication distribution, not just mean present bias). Part A established that commitment demand is a diagnostic for sophistication. Design a field experiment in a savings context whose goal is to measure the distribution of sophistication in the study population — not merely whether a commitment product raises savings. Specify:

- (a) (6 pts) Subject pool and recruitment, and why this population makes the question worth answering.
- (b) (12 pts) Treatments. Your design must include a commitment-device arm with a randomized price (possibly including negative prices, i.e. small bonuses for take-up), and you must explain what the take-up curve as a function of price identifies that a single free-offer arm cannot.
- (c) (9 pts) Elicitation: how you measure each subject's predicted future behavior as well as their actual behavior (the naïf is defined by the gap), and why subjects have an incentive to report honestly.
- (d) (9 pts) Hypotheses, and rough sample-size reasoning: anchor an effect-size guess to Ashraf, Karlan & Yin (28% take-up; savings up ~80%) or to your own Lab 5 simulation, and state the approximate n per arm it implies. Simulation-based reasoning is welcome; formulas are not required.
- (e) (9 pts) The biggest confound — e.g. commitment demand driven by protecting money from household members rather than from one's own future self — and your design answer to it.

Part C: Silicon Pilot Interpretation ⟨AI⟩ (25 points)

Question 4 (The zero-switching-cost puzzle). You run `module5-simulation-lab.py` with one modification permitted by the *CHANGE HERE* blocks: the enrollment form is described as taking zero time — “a single click, right now, on the screen in front of you.” Theory says this should

¹Notation: Fehr & Schmidt (1999) call these parameters α and β ; we write γ for the second to avoid a clash with the discounting β of Question 1.

kill the default effect for every persona: with literally costless switching, even a procrastinator has nothing to postpone. Yet your **exponential** (rational, disciplined) personas still show a 25-point default effect.

- (a) **(8 pts)** Give two mechanisms that could produce this, at least one specific to LLMs (e.g. the model reproducing the famous result rather than responding to the decision problem; defaults read as advice; persona text too weak to override priors).
- (b) **(9 pts)** Design the diagnostic simulation that separates your two mechanisms — what do you change in the prompt or grid, and what result pattern favors which mechanism? Run it if API budget allows and report; otherwise state the predicted patterns precisely.
- (c) **(8 pts)** State the design change a human-subjects default study should adopt as a result of this pilot, and one sentence on what the pilot could not tell you (limits of silicon subjects).

Attach your AI transcripts and, if you ran code, the results CSV.